# Tendon Repair Rehab — Resident Quick Reference

**For use at follow-up visits.** Identify (1) what was repaired, (2) zone, (3) weeks post-op → cross-reference the protocol. Default protocols below follow BWH 2020 / 2007. Surgeon judgment overrides.

## Which protocol?

| Repair | Default protocol |
|---|---|
| FDS and/or FDP, zones 1–5, **4-strand core + epitendinous suture, surgeon clears EAM** | **Flexor EAM** |
| FDS and/or FDP, zones 1–5, **weaker repair, unreliable patient, nerve repair, or surgeon defers EAM** | **Modified Duran** |
| FPL, any zone | **FPL** |
| Extensor (EDC/EIP/EDQ/EPL/ECRL/ECRB/ECU) — any zone | **Primary Extensor** (mallet / boutonniere / zone IV–VIII / SAM) |

**Always confirm**: zone, suture configuration (strands), concomitant pulley repair, nerve repair, vessel repair — these change the plan.

## 1. Flexor (FDS + FDP) — Early Active Motion · Zones 1–5

**Therapy frequency**: 1–2×/week × 8–12 weeks. Orthosis day 3.

| Phase | Orthotic | Exercises | Precautions |
|---|---|---|---|
| **I — Immediate (day 3 to wk 2)** | **Zones 1–3**: dorsal forearm-based; wrist 20° ext, MCP 30–40° flex, IP 0°. **Zones 4–5**: dorsal forearm-based; wrist 0°, MCP 60–75° flex, IP 0°. Add pulley ring if pulley repaired. | Passive DIP & PIP flex, active ext to orthosis. Passive MCP flex 60–80° with active IP ext. Passive wrist flex, active wrist ext to orthosis. **Day 4–5**: after passive, active motion to 25% fist + active ext to orthosis. | No passive wrist ext beyond 0° (zones 4–5) if median/ulnar n. repaired ×6 wk. Avoid place-and-holds (pulley buckling). Tendon weakest day 5–21. |
| **II — Protective (wk 2–4)** | Wk 4: transition orthosis to hand-based. | Active to **50% fist** (wk 2), **75% fist** (wk 3–4). Wk 3: light fine motor out of orthosis in clinic. Wk 4: tendon gliding + isolated FDS glide to repaired finger, wrist 0°–20°; light fine motor at home. | — |
| **III — Intermediate (wk 4–6)** | Gradually wean during day. **Discharge orthosis by 6 wk.** If IPs stiff at 6 wk: convert dorsal block → volar hand-based nighttime (zone 2) or volar forearm-based (zone 3). | Tendon gliding wrist 20–30° ext. **Wk 5**: DIP & PIP blocking. **Wk 6**: progress functional activities. | No composite wrist >30° + combined finger ext until 6 wk. Avoid PIP/DIP blocking of small finger (rupture risk). Weightlifting restricted. |
| **IV — Minimal protection (wk 6–12)** | D/C night orthosis when digital active ext = 0°. | **Wk 8**: light graded strengthening; resisted isolated DIP & PIP flex. Progress to work/sport per MD. | No resisted grip/pinch until 8 wk. |

**Visual cue**: Wk 1–2 = touch IF (25% fist) · Wk 2–3 = touch LF (50%) · Wk 3–4 = touch RF (75%).

## 2. Flexor (FDS + FDP) — Modified Duran · Zones 1–5

**Use when EAM is deferred.** Same orthotic, exercises are passive only in Phase I.

| Phase | Orthotic | Exercises | Precautions |
| --- | --- | --- | --- |
| **I — Immediate (day 3 to wk 2)** | Same as EAM (zones 1–3 vs 4–5). Pulley ring if pulley repaired. | Passive DIP & PIP flex, active ext to orthosis. Passive MCP block 60–80° with active IP ext. **Passive composite fist.** Passive wrist flex, active wrist ext. In clinic (orthosis off): passive wrist ext with fingers passively flexed; passive wrist flex with passive hook fist (prevent intrinsic tightness). | No passive wrist ext beyond 0° (zones 4–5) if nerve repaired ×6 wk. Avoid place-and-holds. Reduced blood flow in zone 1. |
| **II — Protective (wk 2–4)** | Wk 4: hand-based. | Wk 3: light fine motor out of orthosis in clinic. Wk 4: tendon gliding + isolated FDS glide, wrist 0°; light fine motor at home. | — |
| **III — Intermediate (wk 4–6)** | Wean during day. **D/C by 6 wk.** Convert to nighttime orthosis if IP stiffness. | Tendon gliding wrist 20° ext. Wk 5: DIP & PIP blocking. Wk 6: progress function. | No composite wrist >20° + combined ext until 6 wk. Avoid small-finger blocking. |
| **IV — Minimal protection (wk 6–12)** | D/C night orthosis when digital ext = 0°. | Wk 8: light graded strengthening, resisted isolated DIP/PIP flex. | No resisted grip/pinch until 8 wk. |

## 3. Flexor Pollicis Longus — All Zones

| Week | Orthotic | Exercises | Precautions |
|---|---|---|---|
| **0–3** | Dorsal blocking: wrist neutral, CMC flexed/abducted under 2nd MC, MP full ext. **Zone I only**: add separate dorsal gutter blocking IP at 30° flex. | Passive composite thumb flex / active ext to orthotic. Passive IP flex / active ext to orthotic. Gravity-assisted wrist flex / active ext. Tendon gliding digits 2–5. **EAM** (if cleared + 4-strand + epitendinous): place/hold thumb flex with wrist ext, start 48–72 hr post-op. | No active thumb flex unless cleared EAM. No passive wrist ext. No passive thumb ext. No functional use. |
| **3** | Same. | Add place/hold thumb flex with wrist passively extended (gentle). Continue prior. | Avoid co-contraction. |
| **4** | Convert to hand-based. | Active non-resistive thumb flex with wrist ext. | Light prehensile in therapy. |
| **5** | D/C orthosis. | Gentle blocking for thumb IP flex. | — |
| **6** | Dynamic IP ext splinting PRN. | Putty scraping PRN. May start NMES/US PRN. | — |
| **8** | — | Gradually add resistive exercises. | — |

## 4. Primary Extensor — Hand & UE (2020)

### 4a. Mallet (Zone I — DIP) & Zone II (middle phalanx)

| Phase | Orthotic | Exercises | Precautions |
|---|---|---|---|
| **I — Immediate (day 1 to wk 6–8)** | DIP 10°–0° hyperextension. **Tendinous mallet**: 6–8 wk. **Bony mallet**: 6 wk. **Op**: 100% × 6 wk. Provide 2 splints (one for showering). | Active PIP flex of affected finger with adjacent fingers held in extension. | Daily skin check, DIP must stay extended. Consider taping DIP. If swan-neck → reduce passively, dorsal block PIP at 30°. |
| **II — Protective (wk 6 bony / wk 8 tendinous)** | Tendinous: orthosis 100% except exercise/ hygiene. Bony: orthosis for strenuous activity + sleep × 2–4 wk. | Remove orthotic. Gentle AROM DIP flex/ext: start 10° flex, progress 10° per week. Replace orthotic. **Wk 8**: light activity out of orthosis if no lag. | If DIP extensor lag ≥10° → orthosis 100% × 2–4 wk. |
| **III — Intermediate (wk 10)** | D/C during day. Night orthosis × 2 more wk. | Fine motor, gradual flexion while maintaining DIP ext. | Most zone 1–2 injuries finish with –10–0° extensor lag. |

### 4b. Boutonniere (Zone III — PIP)

| Phase | Orthotic | Exercises | Precautions |
|---|---|---|---|
| **I — Immediate (day 1 to wk 6)** | Orthosis or circumferential cast with PIP at 0°. Op: 100% × 6 wk. | Active DIP flex of affected finger. | If DIP hyperextension at wk 2 → reduce passively. |
| **II — Protective (wk 6)** | Convert to orthosis with PIP at 0°. Wk 7: reduce gradually as 0° PIP ext maintained. Light activity if no lag. | Gentle active PIP ext to 30° flex, progress 10° per week. Replace orthotic. | If PIP extensor lag ≥10° → orthosis 100% × 2–4 wk. |
| **III — Intermediate (wk 10)** | D/C. | — | — |

### 4c. Zones IV–VIII (proximal phalanx through forearm) — Immobilization Protocol

| Week | Orthotic | Exercises | Precautions |
|---|---|---|---|
| **1–6** | Volar digit static, PIP at absolute 0°, or serial cast. **Lateral bands repaired** → include DIP at 0°. **Lateral bands intact** → DIP free. | ROM may start wk 3–6 depending on healing. AROM PIP flex to 30°, progress 10–20°/wk if no lag, 10 reps hourly. Lateral bands repaired → tendon gliding wk 3. Intact → wk 1. | No forceful flex. No gripping. Splint between sessions. Serial cast if PIP contracture, closed injury, or noncompliant. |
| **6–8** | Wean from splint during day. Night splint continues. | AAROM or dynamic flexion splint may start. Composite wrist + digit flex. | Light functional use out of splint (≤1–3 lb). |
| **10–12** | D/C splint. | PROM / PREs. | — |

### 4d. Zones III–IV Alternative — Short Arc Motion (SAM) Protocol

*Consider when broad tendon-bone interface in zone IV and adhesion risk is high.*

| Week | Orthotic | Exercises | Precautions |
| --- | --- | --- | --- |
| **1** | Digit volar immobilization: PIP & DIP at 0°, 24/7 except exercise. Two volar **exercise** splints: **template 1** (PIP 30° flex, DIP 20° flex), **template 2** (PIP 0°, DIP free). | Remove immobilization hourly, AROM PIP & DIP, 10–20 reps in templates 1 & 2. Wrist held 30° flex, MP at 0°. Lateral bands repaired → limit DIP flex to 30–35° in template 2; intact → fully flex/extend DIP. | PIP must be at 0° in immobilization splint to prevent extensor lag. |
| **2** | If no lag: progress **template 1** → PIP 40–50°, DIP 30–40°. | If extensor lag develops → smaller flex increments, focus on extension. | If rupture suspected → MD. |
| **3** | If no lag: **template 1** → PIP 50–60°, DIP 40–50°. | — | — |
| **4** | If no lag: **template 1** → PIP 70–80°, DIP 50–60°. | — | If PIP stiff → intermittent splint into flex, continue static ext splint into wk 5–6. |
| **5** | Begin weaning. | Composite flex and gentle PREs. | Light functional activities out of splint. |
| **6** | D/C splint. Night PRN. | PROM & PREs, reverse putty scraping. | — |

## Red flags at follow-up

- **Sudden loss of active flexion / extension** after gain → assume rupture. Refer to surgeon.
- **Extensor lag ≥10°** → return to full-time orthosis × 2–4 weeks.
- **PIP contracture developing** → serial casting or static progressive splint, escalate therapy.
- **No improvement in tendon glide after 6–8 weeks** → discuss tenolysis (typically deferred to ≥3–6 months post-op).
- **Persistent edema beyond 4 weeks** → edema control, rule out infection.

---

## Key principles

- Repair is weakest **day 5–21**.
- Strong repair (4-strand core + epitendinous) is the prerequisite for EAM.
- Adhesions favor extrinsic healing; controlled motion favors intrinsic healing.
- Gap >3 mm → ↑ rupture and adhesion risk.
- Always cross-check with surgeon's op note (suture configuration, pulley/nerve/vessel work).

## Sources (BWH Department of Rehabilitation Services)


- Flexor EAM (2020) — https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/flexor-tendon-repair-hand-therapy-protocol.pdf
- Modified Duran (2020) — https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/modified-duran-hand-therapy-protocol.pdf
- FPL (2007) — https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/hand-fpl-repair-pt-protocol-bwh.pdf
- Primary Extensor (2020) — https://www.brighamandwomens.org/assets/BWH/patients-and-families/pdfs/upper-extremity-extensor-tendon-repair-protocol-2020.pdf

BWH protocol index: https://www.brighamandwomens.org/patients-and-families/rehabilitation-services/physical-therapy-protocols


*Compiled by Dr. Janos Barrera (NYU Langone) as a teaching aid. Always defer to the operating surgeon's specific instructions.*